



Embedding-Based Multilingual Semantic Search for Geo-Textual Data in Urban Studies

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Abstract

Urban studies increasingly rely on vast amounts of geo-textual data from social media and news archives. However, effectively searching and analyzing this data remains a significant challenge. Current keyword-based search methods struggle to capture the inherent vagueness, multilingualism, and dynamic nature of this information, thus limiting comprehensive urban analysis and insight discovery. To address this, we propose a novel and modular geospatial semantic search (GSS) workflow. This framework leverages multilingual text embeddings to enable effective semantic search across flexible geographical units, moving beyond the limitations of keyword matching. The GSS workflow integrates pre-computed document embeddings, adaptive spatial aggregation methods, and interactive map visualization to facilitate the exploration of search results. Our approach enables efficient semantic exploration of user-contributed mass data, improving the understanding of public expression in urban areas, monitoring thematic trends and localized events, and offering a scalable and privacy-aware design. We demonstrate these capabilities in a proof-of-concept using hexagonal bins for visualization.

Keywords Semantic similarity · Geo-textual data · Text embeddings · Urban studies · Social media

Introduction

The analysis of large volumes of human-generated geo-textual data, derived from diverse sources such as social media, news archives, and Wikipedia, is becoming increasingly important for understanding urban dynamics and addressing socio-environmental challenges (Ilieva and McPhearson 2018; Schwering 2008). However, a critical bottleneck exists in an effective search approach that exploits the rich information contained within these data. The inherent ambiguity and vagueness of natural language pose significant challenges to the extraction of meaningful insights.

Traditional geospatial search workflows, which rely heavily on full-text search using analyst-selected keywords (Hervey et al. 2020), are proving inadequate for handling the semantic richness of user-generated content. These keyword-based methods are particularly ineffective in analyzing the dynamic nature of everyday language, which includes creative expressions, rapid evolution (e.g., neologisms, “Internet language,” slang), multilingual compositions, phonetic variations, and occasional errors such as misspellings that are common in social media (Sundaram et al. 2023). The reliance on exact keyword matching severely limits the ability of users of these systems to discover relevant information and conduct comprehensive analyses. This is particularly true for analyses of urban systems conducted through the lens of geosocial media. These systems typically span many different contexts and require recursive spatial, temporal, and topical exploration (Dunkel et al. 2019).

Semantic similarity, which quantifies the degree of semantic relatedness between documents or terms, offers a compelling alternative to these lexical methods for geo-textual search (Khem et al. 2023). By focusing on meaning rather than exact word matches, it allows for improved query, comparison, and ranking tasks. This provides the basis for overcoming the limitations of keyword-based search and

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significantly improving the effectiveness of information retrieval. Such an algorithmic shift is particularly important for applications in urban studies, where understanding the public perceptions of a city requires the analysis of multiple sources, where different languages and idioms are common, often leading to errors in semantic matching routines. Semantic search promises to unlock deeper insights from complex geo-textual datasets.

The mapping of geospatial semantics is also crucial for the spatial analysis of geo-social data, and interactive map visualization becomes essential for making semantic search results accessible and actionable. Such geo-textual data, including social media posts, Wikipedia entries, news articles, and volunteered geographic information (VGI) data, has become a critical resource for addressing various socio-environmental challenges. These data sources have the potential to improve our understanding of public needs in urban neighborhoods, monitor negative social trends, and support public engagement in urban planning (McKittrick et al. 2023; Ilieva and McPhearson 2018). However, without effective search and visualization tools, the full potential of this geo-textual data remains untapped. Improved methods for categorizing, integrating, and exploring user-generated geo-textual data are essential, and visualization plays a key role in this exploration process.

Therefore, the core problem addressed in this work is the limited effectiveness of current search methods in extracting meaningful insights from the vast and semantically rich landscape of multilingual geo-textual data in urban studies. Keyword-based approaches are insufficient to capture the nuances of language and meaning in these datasets, hindering comprehensive analysis. To overcome this challenge, this paper proposes a novel and scalable geospatial semantic search (GSS) workflow.

In presenting this contribution, the paper first outlines the general, modular architectural pipeline of the GSS workflow, then discusses specific configuration approaches for its core components, and finally illustrates its practical application and evaluation through a detailed case study; this structure allows the paper to move from broadly transferable principles to increasingly concrete implementation details and empirical findings. Leveraging recent advances in multilingual text embeddings, the GSS workflow provides an efficient and robust solution for semantic search across flexible geographic units, moving beyond the constraints of lexical matching. Crucially, GSS incorporates interactive map visualization as an integral component. This allows users to spatially explore and interpret semantically relevant text, which improves search accessibility, quality, and overall system performance in geospatial contexts. GSS provides a practical and user-friendly way to unlock the wealth of semantic information hidden within urban geo-textual data, making it a valuable tool for urban studies.

Related Work

This section explores the existing body of research relevant to our proposed GSS workflow. We first examine the landscape of geo-textual data querying, focusing on keyword-based approaches and their limitations, which motivate the need for semantic search. Subsequently, we highlight the advancements in semantic similarity measures and their application within GIScience.

Geospatial Full-Text Search (GFTS)

Early approaches to querying geo-textual data relied heavily on keyword-based search techniques. These techniques, often described under the umbrella term GFTS, use spatial location data and associated textual descriptions to retrieve relevant documents (Cong and Jensen 2016; Chen et al. 2021). Keyword-based queries, in which analysts manually select terms, are effective for targeted information retrieval when the vocabulary is well defined, and the user queries accurately match the document lexicon (Hervey et al. 2020). For example, point-of-interest (POI) exploration, a key task in this domain, often uses keyword queries to map businesses or landmarks based on names or categories (Cao et al. 2014; Chen et al. 2021). These methods are fundamental and have undergone extensive development, including techniques for optimizing spatial indexes, improving query accuracy, and handling complex spatial relationships (Cao et al. 2014; Chen et al. 2021, 2019).

Keyword-Based Geo-Textual Search

Chen et al. (2021) provide a comprehensive overview of geo-textual data retrieval techniques, including contemporary keyword-based geo-textual search. They categorize the methods into static and streaming data. For static queries, they highlight basic techniques such as identifying POIs within areas of interest (AOIs). More complex queries include temporal and social context. For streaming data, they classify approaches into publish/subscribe systems, localized event detection, temporal-spatial keyword queries, and location-based term queries. This highlights the challenges of real-time processing of high-volume data streams. In a previous study, Chen et al. (2019) further categorized spatial keyword retrieval studies based on input data, output results, and methodologies. This can be seen as a structured overview of the development of the field. In their outlook, managing high-dimensional geo-textual data and mining latent relationships in large datasets are highlighted areas for further research.

Cong et al. (2016) specifically examine geo-textual data exploration for social media platforms like Twitter, focusing on region search and region exploration. They propose query personalization based on user histories to improve relevance. Liu et al. (2020) address search engine optimization (SEO) for geo-textual data, emphasizing keyword selection for maximizing a location's influence zone, a critical aspect for businesses seeking online visibility within geospatial datasets. These studies collectively illustrate the breadth and depth of research dedicated to refining keyword-based geo-textual search, optimizing indices, and adapting techniques to diverse data types and user needs. A common thread throughout this research is the fundamental role of keyword selection and management in effective geo-textual information retrieval (Chen et al. 2021, 2019; Liu et al. 2020).

Limitations of Keyword-Based Search

Despite advancements, keyword-based geo-textual search retains inherent limitations that restrict its effectiveness, particularly when dealing with user-generated content, lexical variation, and nuanced semantic queries (Lee et al. 2018; Yao et al. 2010). These methods rely heavily on exact or near-exact lexical matching, struggling with the ambiguity and vagueness of natural language, including synonyms, polysemy, and diverse linguistic styles prevalent in sources like social media (Hu 2018; Sundaram et al. 2023). Users must anticipate the precise keywords used in documents, a significant challenge given the dynamic and informal nature of online language, which often encompasses slang, neologisms, and multilingual expressions (Sundaram et al. 2023; Lee et al. 2018).

Furthermore, keyword-based searches are brittle in the face of typographical errors and phonetic variations, common in user-generated content (Yao et al. 2010; Li et al. 2016). Workarounds such as different versions of Levenshtein distances (Castells-Rufas 2023) or curated vocabularies attempt to address this lexical variation but remain inherently limited. They often require expert curation by domain specialists or native speakers, struggle to capture subtle semantic nuances, remain prone to errors, and ultimately hinder efficient search and exploration of large and diverse datasets (Lee et al. 2018). Even with iterative query refinement or vocabulary curation, which adds complexity and requires manual intervention (Chen et al. 2021; Lee et al. 2018), the fundamental reliance on keywords limits the ability to identify relevant information based on meaning.

This results in missed spatio-social insights and incomplete understanding of urban phenomena, highlighting the need for methods that move beyond simple lexical matching (Lee et al. 2018; Khem et al. 2023). This limitation is not unique to geo-textual search; in the field of spatial

co-location pattern mining, for instance, traditional methods based on co-occurrence frequency are similarly criticized for overlooking rich contextual information, which can lead to insufficient accuracy and spurious results (Zhou et al. 2025).

Deep Learning for Enhanced Geo-Textual Analysis

Recent research explores deep learning to overcome the limitations of keyword-based methods in geo-textual analysis (Berragan et al. 2023; Wu et al. 2023; Wang et al. 2024a; Ghosal and Jain 2023; Schmidt et al. 2023). Deep learning techniques are applied to sentiment analysis, topic modeling, and named entity recognition (NER) within geospatial contexts, leveraging the ability of neural networks to capture complex patterns and semantic nuances (Ghosal and Jain 2023). Geospatial sentiment analysis, for example, uses deep learning models to identify spatial patterns of sentiment in geo-referenced texts, offering insights into public opinions and emotional responses across geographic areas (Schmidt et al. 2023). Geospatial topic modeling employs deep learning to identify prevalent themes within specific locations, aiding in understanding the thematic characteristics of different regions (Wang et al. 2024a). Deep learning-based NER is used to extract and classify geospatial entities from unstructured text, converting fuzzy location references into machine-readable geometric entities (Berragan et al. 2023; Wu et al. 2023). Transformer-based models, such as DistilBERT and RoBERTa, have shown promise in NER and sentiment analysis of geo-textual data, demonstrating the effectiveness of deep learning in enhancing the analytical capabilities beyond keyword-based approaches (Berragan et al. 2023; Schmidt et al. 2023).

Despite these advances in applying deep learning to specific geo-textual analysis tasks, the integration of semantic similarity, enabled by text embeddings from deep learning models, into interactive map-based exploration and search workflows remains an open area (Mai et al. 2020). Specifically, techniques for document aggregation and ranking based on text embedding similarity for map-based visualizations are still under development (Mai et al. 2020). Our work addresses this gap by introducing a GSS workflow that leverages multilingual text embeddings and flexible aggregation techniques to enable interactive map-based visualization of semantically relevant texts, moving beyond the constraints of keyword-based searches and exploring the potential of semantic similarity in geospatial contexts.

Semantic Similarity

Semantic similarity offers a powerful alternative to keyword-based search by quantifying the meaning-based relatedness between texts, leveraging advancements in natural language processing (NLP) and deep learning (Khem et al.

2023; Greiner-Petter et al. 2020). By focusing on semantic resemblance rather than exact word matches, semantic similarity enables more effective information retrieval, particularly in scenarios involving diverse language use, ambiguity, and evolving terminology (Lee et al. 2018).

Advances in Text Embeddings

The foundation of semantic similarity lies in text embeddings, numerical vector representations of text that capture contextual meaning (Greiner-Petter et al. 2020). Early advancements in word representation were driven by models like Word2Vec (Mikolov et al. 2013), GloVe (Pennington et al. 2014), and FastText (Bojanowski et al. 2017), which generate static vector representations. While effective for certain tasks, their inability to fully capture contextual variations was a key limitation. This limitation was overcome by the advent of transformer-based models (Vaswani et al. 2017), such as BERT and its variants. By employing attention mechanisms to dynamically incorporate context, these models have demonstrated superior performance and established new state-of-the-art results for most NLP tasks (Vaswani et al. 2017; Chen et al. 2024).

Multilingual embedding models, like Multilingual E5 and M3-Embedding, further extend the applicability of semantic similarity by enabling cross-lingual information retrieval (CLIR) and handling multilingual datasets, critical for global applications and diverse user-generated content (Wang et al. 2024b; Chen et al. 2024; Litschko et al. 2022). M3-Embedding, for instance, generates both sparse and dense vector representations for text and images, facilitating hybrid retrieval across modalities (Chen et al. 2024). Multilingual E5 achieves robust semantic representations across languages through contrastive pre-training on massive multilingual text pairs, making it particularly suitable for urban studies analyzing diverse, multilingual geo-textual data (Wang et al. 2024b).

Key to the power of these modern embeddings is their pre-training on vast, diverse internet-scale corpora like Reddit, which includes the slang, neologisms, and misspellings that challenge lexical methods. Moreover, they typically employ subword tokenization strategies. Such strategies, proven effective for fuzzy matching and handling out-of-vocabulary terms in production systems like Facebook search, grant structural robustness when encountering novel or varied word forms (Huang et al. 2020). These advancements in text embeddings provide the necessary tools to operationalize semantic similarity for enhanced geo-textual data analysis and search.

Semantic Similarity in Information Retrieval and GIScience

Information retrieval using semantic similarity typically involves chunking documents, embedding chunks, and retrieving relevant chunks based on similarity to a query (Luo et al. 2024). Chunking is necessary for long documents exceeding the embedding model's context window (Luo et al. 2024). Document embeddings can be created by averaging chunk embeddings, balancing precision and storage efficiency (Seegmiller et al. 2023). Retrieval is performed using distance functions like cosine distance to identify semantically similar chunks (Luo et al. 2024).

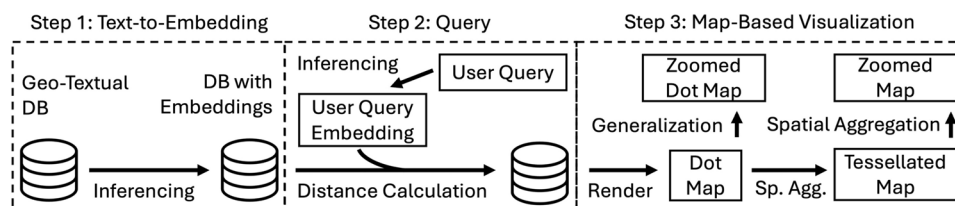
Within GIScience, semantic similarity has been recognized for its potential in geospatial search for over 15 years, with early frameworks proposed for measuring semantic similarity in geographic information retrieval (Janowicz et al. 2011; Schwering 2008). Janowicz et al. (2011) laid theoretical groundwork for semantic similarity measurement in GIScience, providing a framework for comparing and selecting measures across applications. Miao et al. (2021) developed the OGCDSSM model, integrating spatiotemporal and thematic similarity using ontology semantics, demonstrating improved search performance over keyword-based methods, though its reliance on ontologies presents practical challenges. Mai et al. (2020) proposed a semantic query expansion framework for geoportals using GloVe embeddings, focusing on ranking top documents rather than interactive map-based search.

Despite these contributions, the effective operationalization of semantic similarity in interactive geospatial search, particularly leveraging recent advancements in transformer-based embeddings for map-based visualization, remains a largely unexplored area, which our research aims to address. Our proposed GSS workflow builds upon this established foundation, integrating state-of-the-art multilingual text embeddings and interactive map visualization to provide a practical and scalable solution for geo-textual data exploration and search.

Geospatial Semantic Search (GSS)

Geospatial semantic similarity, as we define it, quantifies the degree of correspondence for documents of a geo-textual database with both textual content and location information. GSS, in turn, refers to the user-driven process of employing semantic search to locate and rank similar documents to a user query on a map. While GSS is a subset of geospatial

Fig. 1 Geospatial semantic search workflow consisting of three steps: text-to-embedding database preprocessing, query, and map-based visualization



semantic similarity, the latter has applications beyond search, such as analyzing the thematic coherence of geographic areas without direct user queries.

This section introduces the GSS workflow by distinguishing between its sequential *steps* and its foundational *modules*. *Steps* refer to the three main sequential phases guiding the practical design of real-world applications. *Modules*, conversely, represent the underlying core architectural units from a conceptual standpoint. While the three *steps* are inherent to GSS, its *modules*, such as the underlying embedding model, can be replaced and updated as research in NLP and urban studies progresses.

GSS Steps

Step 1 involves preparing a geo-textual database (Figs. 1 and 2). Each document in this database must contain both a textual component and a geospatial reference, such as a point, line, or polygon. Common data sources include social media platforms like Instagram, Flickr, Facebook, TikTok, and X (formerly Twitter), as well as news articles, Wikipedia, and OpenStreet-Map. This step includes tasks such as stripping out personally identifiable information (PII) to ensure privacy compliance, data cleansing to improve data quality, and deduplication to remove redundant entries or merging contextual metadata (titles, comments, geoaccuracy-related preprocessing).

For each document (e.g., a social media post with its caption), only the textual component is transformed into a fixed-length vector representation using a pre-trained embedding

model; this process is commonly known as inferencing (Reimers and Gurevych 2019). The embedding model represents the core module of GSS as the workflow directly inherits the intrinsic properties of the selected model, including its multilingual capabilities, overall performance—particularly crucial for cross-lingual information retrieval (CLIR)—and any inherent biases (“Challenges and Limitations” section). The embeddings generated by the chosen model are used without modification for the subsequent steps. Consequently, the effectiveness of GSS is heavily dependent upon the quality and characteristics of the chosen embedding model (“Performance and Scalability” section).

The model-agnostic design ensures that as advancements in embedding models emerge (e.g., specialized models for social media, underrepresented languages, or multimodal data like geo-visual information), they can directly benefit the existing GSS workflow by simply repeating the inferencing step once (Fig. 2) without fundamental changes to the overall system architecture. The geospatial reference, or location, itself remains unchanged and is simply associated with this newly generated text embedding.

Consequently, each document, initially defined by its text and location, is now represented by its original location, paired with its corresponding embedding. If subsequent analyses use data adhering to the same database format (location column and embeddings from the same model), diverse preprocessed databases can be used in a plug-and-play manner, or they can be merged into a single database without needing to adapt any other steps or modules.

Original Documents		Embeddings		Lat		Lon	
Coffee break ☕	$f(\text{text})$	$[v_1, v_2, \dots, v_n]$	①	50.234			7.101
#Cherryblossom 🌸	$f(\text{text})$	$[v_1, v_2, \dots, v_n]$	②	50.123			7.203
Spring is here 🌸	$f(\text{text})$	$[v_1, v_2, \dots, v_n]$	...	50.832			7.423
@Ferrari 4 the win!	$f(\text{text})$	$[v_1, v_2, \dots, v_n]$	④	50.231			7.233

Fig. 2 Step 1—Text preprocessing of the geo-textual DB with text inferencing. Note that the location (here coordinates) is preserved in its original format and is not input to the embedding model

Step 2 is triggered by a user query (Fig. 1). The query undergoes the same embedding process as the database texts, utilizing the same embedding model module selected previously. Subsequently, the query embedding is compared to each document embedding in the database using a distance function, such as cosine distance, which acts as a critical submodule linked to the embedding model. While embedding models are typically optimized for use with a specific distance function (e.g., cosine distance/similarity, which is preferred for most state-of-the-art embedding models), the modular design allows for the exploration of alternative distance functions depending on the specific application and desired similarity metric. Consulting the documentation provided by the embedding model developers is recommended for selecting the appropriate function.

Each document receives a similarity score based on this comparison, where higher scores signify greater relevance to the query and lower scores indicate lower relevance. These scores, which can vary significantly across different models, are often normalized to a 0–1 range. The top-ranked documents, after being sorted by descending similarity score, are considered the most relevant matches to the user's query. Optionally, further submodules can be incorporated at this stage to refine the embedding retrieval process. For instance, hybrid approaches combining semantic similarity with traditional lexical methods like BM25 could be implemented to leverage the strengths of both paradigms (e.g., BM25 retrieving candidates, followed by semantic reranking). Alternatively, dedicated reranking models, trained to optimize relevance based on semantic understanding, could be integrated as submodules to further enhance the precision of the subset of top k results, although the primary GSS visualization typically utilizes all scores.

Step 3 focuses on visualizing the results (Figs. 1 and 3). In contrast to common retrieval tasks in retrieval-augmented generation (RAG) workflows, which typically return only the top k results, GSS uses all similarity scores for map-based visualizations.

For basic applications, this map module can display lightness-based points (Fig. 3a) representing data using variations in luminance, commonly employing a gray-scale or another suitable hue. Higher similarity scores are encoded with *lower* lightness values (darker grays), while lower scores correspond to *higher* lightness values (lighter grays). Radius-based point maps (Fig. 3b) encode similarity scores directly as point radii, often scaled by a constant factor. Optionally, within the map module, point radii can be used to visualize an additional attribute, such as the overall number of posts associated with a location (Fig. 4).

However, for larger urban datasets, leaving the micro-scale, more sophisticated scenarios require spatial aggregation usually in the form of regular tessellation as raster-based (Fig. 3c) or hexagon-based tessellation (Woźniak and Szymański 2021). This allows for the identification of urban hotspots or the representation of continuous semantic urban areas like land use patterns (e.g., business- or leisure-driven areas). Spatial aggregation functions (SAFs) are required for aggregating the multiple similarity scores falling within each tessellation unit (e.g., hexagon) into a single representative value (Fig. 3c, detailed in the “Embedding Aggregation” section, and Table 2). For instance, the average similarity score within a tessellation geometry can be used as input to a luminance function (Fig. 3c) or more general color schemes.

This represents the fundamental concept of GSS. As previously mentioned, the three steps and their underlying modules require careful tuning to achieve reliable and performant results.

Embedding Operations for Geo-Textual Data

Understanding the characteristics of high-dimensional embeddings is essential for choosing subsequent processing steps for visualizing similarity within the GSS workflow. Several mathematical operations commonly employed in NLP are applicable for embedding aggregation, including

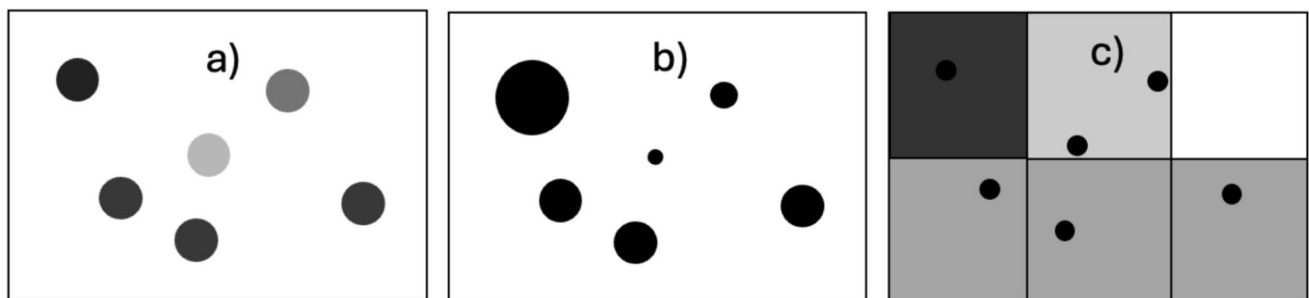


Fig. 3 Step 3—Common map-based visualization types for point values using: **a** luminance, **b** radius, **c** aggregation by tessellation (e.g., average of all points in a raster cell)

vector arithmetic and distance calculations. These operations and their relevance for GSS are shown in Table 1.

Vector operations, such as addition and subtraction, generate new embeddings that roughly reflect the operation's semantic meaning. Addition combines the meanings of two embeddings, while subtraction can emphasize their differences. Scalar multiplication modifies an embedding's magnitude without altering its direction, useful for adjusting its relative importance. However, scalar multiplication is not compatible with cosine distance, which only considers directionality. A single embedding can be created by averaging individual embeddings, with or without weights, or by calculating their median (Seegmiller et al. 2023). Averaging a large number of embeddings leads to a loss of information granularity (Zhu et al. 2024) but reduces memory requirements. While not a complete privacy solution, averaging embeddings can increase the difficulty of reconstructing the individual vectors (Morris et al. 2023).

Embedding Aggregation

The GSS workflow incorporates several optional embedding aggregation stages to manage data complexity and tailor the analysis to specific research questions. These stages address different dimensions of data granularity, impacting the level of detail preserved in the semantic representations.

Granularity, in this context, refers to the level of detail or specificity in the data representation. Finer granularity implies a more detailed representation but also higher storage requirements, while coarser granularity represents data in a more generalized or summarized manner for drastically

reduced storage requirements (Table 3). Each aggregation stage involves partitioning the data based on specific criteria, resulting in distinct levels of granularity.

Drawing upon the conceptual framework proposed by Dunkel et al. (2019) for analyzing collective reactions in location-based social media, we can structure our approach to embedding aggregation along four key facets: *thematic*, *spatial*, *temporal*, and *social*. While Dunkel et al. focused on reactions to events, their multi-faceted framework provides a structure for organizing the aggregation of any geo-referenced textual data by considering what, where, when, and who is being communicated.

The aggregation of embeddings itself typically occurs as a preprocessing step (Step 1) to create representative embeddings for these partitions (e.g., an average embedding for all documents within a specific geographic area and/or time window). These representative, aggregated embeddings can then be stored and used in Step 2 for comparison against the query embedding, as exemplified by the location-averaged embeddings in our use case ("[Performance and Scalability](#)" section). Key aggregation strategies within this framework include the following:

Thematic Aggregation

For long documents, initial chunking into sentences or paragraphs creates document partitions and fine-grained, chunk-level embeddings. Aggregating these chunk embeddings (e.g., averaging) yields a single, coarser document-level embedding. Choosing between chunk-level or document-level embeddings depends on the desired semantic detail

Table 1 Vector operations and distance calculations relevant for geospatial semantic search

Category: operation	Description	Use cases
Vector operation: addition and subtraction	Combines two embeddings or highlights their differences	Merging or comparing semantic content of text embeddings
Vector operation: scalar multiplication	Alters the magnitude of an embedding without changing direction. Not to be used in combination with cosine distance	Adjusting the significance of a particular embedding
Vector operation: average vectors (weighted/unweighted)	Combines multiple embeddings into a single representative vector	Aggregating semantic content from various sources. Necessary for memory reduction and faster spatial aggregation
Vector operation: median vectors	Determines the median embedding to minimize the sum of absolute differences in a set	Finding a central vector representative of a set without extreme value influence. Alternative to averaging vectors
Distance calculation: dot product	Assesses similarity by summing products of corresponding elements	Capturing alignment between vector components for similarity searching and clustering
Distance calculation: cosine similarity	Computes cosine of the angle between vectors, emphasizing orientation over magnitude. Preferred distance function for most embedding models	High-dimensional data analysis where directionality is more significant than magnitude
Distance calculation: Euclidean distance	Measures straight-line distance between points, reflecting physical separation	Straightforward measure of distance for clustering and classification tasks

versus computational efficiency. Apart from this intra-document aggregation, inter-document aggregation can aggregate documents by shared themes for instance based on tags, keywords, categories or sentiments.

Spatial Aggregation

Documents are grouped based on their *original* geospatial references (e.g., points falling within specific hexagonal bins, or documents whose polygonal geometries intersect with pre-defined administrative areas), forming spatial units. Aggregating the text embeddings within a single spatial unit produces a spatially aggregated embedding, representing the collective semantics of that unit at a coarser spatial granularity. Smaller spatial units offer finer spatial detail but may have sparser data; larger units generalize semantic patterns over broader areas. The choice of spatial unit defines the spatial granularity.

Temporal Aggregation

Similarly, documents for the same spatial unit can be partitioned by time intervals (e.g., all posts from a specific hour, day, month, or year) or recurring patterns (e.g., specific events, seasons). The timestamp associated with each document determines its assignment to a temporal partition. The text embeddings of documents within each temporal partition are then aggregated (e.g., by averaging their vectors) to create a temporally aggregated embedding. These aggregated embeddings can then reveal semantic trends over different time periods at a coarser temporal granularity.

Social Aggregation

Datasets can be partitioned by user metadata (e.g., user type, user ID, demographics, platform). Aggregating embeddings within these partitions yields social group-aggregated embeddings, summarizing semantics for each category at a coarser granularity of social communities. This reduces data size but sacrifices some social-semantic detail.

The sequence of aggregation steps is critical, as it dictates the final analysis structure and granularity. Different orders yield different summaries because aggregation inherently loses some detail. For example, temporal aggregation followed by spatial preserves temporal variations within larger regions, while the reverse order averages them out. Careful consideration of the desired granularity and the appropriate sequence of aggregation steps is essential.

Usability Considerations: from User Query to Spatial Visualization

After the documents have been aggregated to a desired level during preprocessing (Step 1, Figs. 1 and 2), a user query

(e.g., “Best vegetarian restaurant”) initiates the actual visualization process. The query is embedded and compared to all database embeddings (Step 2), yielding numeric similarity scores for each database entry. A key challenge arises when moving from these similarity scores to map-based visualization, particularly when multiple documents with different scores relate to the same spatial unit. For instance, if multiple documents reference the same coordinate, or when using spatial tessellation where cells contain numerous documents, a method is needed to derive a single score within each spatial unit.

To address this, our GSS workflow incorporates spatial aggregation functions (SAFs). It is important to distinguish these SAFs, which operate on-the-fly on the similarity scores for visualization, from the embedding aggregation strategies discussed in the “Embedding Aggregation” section, which are part of the database preprocessing (Step 1). SAFs serve to consolidate multiple similarity scores—not embeddings—within each spatial unit into a single value for display. SAFs serve to determine this single, representative similarity value for each spatial unit (point, line, polygon) based on the underlying document similarity scores. This process breaks down into two primary methodological components:

- 1) SAF selection: Choosing an appropriate SAF to aggregate similarity scores within each spatial unit. This function determines how multiple similarity scores are combined into a single value for visualization. Common options include maximum, mean, minimum, median, range, and variance.
- 2) Color function definition: Defining a color function to map the aggregated similarity values (output from the SAF) to a visual color scale on the map. This function determines how different similarity values are represented visually.

In designing the SAF component, we initially considered using the mean function for aggregation. The rationale was that averaging similarity scores within a spatial unit might provide a generalized representation of relevance across varying zoom levels. Instead, we recognized a potential challenge: the risk of what we term “value dilution.” This phenomenon, where strong individual semantic signals are diminished when averaged over larger areas containing many weaker or irrelevant signals, is a direct manifestation of the scale effect within the well-known modifiable areal unit problem (MAUP) (Openshaw 1984), with recent studies continuing to explore the impact of tessellation shape and scale on urban analysis (Monzur et al. 2024). The MAUP highlights that the results of spatial analysis can vary significantly depending on the size (scale) and shape (zoning) of the areal units used for aggregation. Indeed, selecting an

Table 2 Advantages, disadvantages, use cases, and examples for spatial aggregation functions

SAF	Advantages (+); disadvantages (-)	Use case	Example
Min	+ Highlights the lowest value clearly - Can be misleading if outliers are present - Disregards the quantity of low values - Effective only for high zoom levels	Identifying areas in total absence of a certain document or finding “pure” areas where even the lowest score is highly relevant	Geographies of fear: looking for a “safe, peaceful and quite” area as path home. The minimum score would point out places to avoid in each bin
Max	+ Emphasizes highest value + Good for search + Effective for low and high zoom levels - Can be misleading if outliers are present - Disregards the quantity of high values	Identifying areas with at least one certain document, suitable for classic search cases but is not accurately representing the area	“Tourism hotspot” detection: highlighting areas with at least one extremely popular landmark or tourist attraction, regardless of other surrounding lower-scoring locations
Mean	+ Good for an overall representation + Visually represents underlying trend - Can be misleading if outliers are present - Value dilution	General distribution analysis, suitable for representing the trend in an area	Analyzing overall relevance of documents related to “historic landmarks” across a city, providing a sense of where mentions are most concentrated
Median	+ Good for an overall representation + Not affected by extreme values, more robust than mean + Visually represents underlying trend - May not reflect well the actual distribution - Value dilution	Analyzing central tendency, suitable for representing the trend in an area	Determining central relevance in regions for queries such as “eco-friendly urban spaces,” eliminating influence from extreme high or low scores
Range (max–min)	+ Highlights the diversity of scores through the difference between the highest and lowest score - Can be misleading if outliers are present - Value dilution	Analyzing spatial clusters with a similar distribution for low or high diversity	Comparing areas for diversity of relevance in documents mentioning “local cuisine,” identifying clusters with either consistently high or low scores. Combined with a minimum threshold, only positive clusters can be filtered
Variance	+ Highlights the diversity of scores - Ignores the absolute values - Disregards the quantity of low values	Analyzing spatial clusters with a similar distribution for low or high diversity	Comparing areas for diversity of relevance in documents mentioning “local cuisine,” identifying clusters with either consistently high or low scores. Combined with a minimum threshold, only positive clusters can be filtered

optimal static spatial granularity for social media analysis is a complex task, often requiring multi-criteria approaches to balance conflicting objectives, such as maximizing spatial autocorrelation while minimizing local variance (de Andrade et al. 2020).

However, our GSS workflow is primarily designed for interactive exploration, where users dynamically change the effective spatial resolution by zooming. Therefore, rather than seeking a single optimal static grid size, we focused on SAFs that provide consistent insights across these varying scales. The “value dilution” issue associated with MAUP’s scale effect is particularly problematic for functions like mean in an interactive context: at lower zoom levels (larger effective bins), highly relevant documents could be averaged out, leading to an underrepresentation of truly relevant areas.

Conversely, at higher zoom levels, where spatial units are smaller and contain fewer documents, the dilution effect of the mean would be less pronounced, potentially leading to inconsistent visual representations across zoom levels. Furthermore, the mean function can be influenced by varying document densities across the dataset, potentially introducing spatial biases, especially at dataset boundaries.

To mitigate the potential MAUP-related “value dilution” effect and ensure consistent hotspot identification during interactive zooming, we systematically evaluated a range of SAFs. We hypothesized that different SAFs would be suited to different user needs and query types, particularly concerning their robustness to changes in effective spatial resolution during interactive map zooming. Table 2 summarizes the characteristics of the SAFs we considered. Starting with the mean as a baseline, we explored maximum, minimum, median, range, and variance functions, analyzing their advantages, disadvantages, and potential use cases within the GSS workflow. Our analysis, detailed in Table 2, aimed to identify the most effective SAFs for different GSS scenarios. We anticipated that the choice of SAF would significantly impact the interpretability and usability of the resulting map visualizations, and therefore, careful consideration of SAF selection is a crucial aspect of the GSS methodology.

Finally, after determining the aggregated similarity values via a chosen SAF, the visualization process requires a color encoding scheme to translate these numerical values into a visually interpretable map. While color mapping principles are well-established for cartographic visualization, the selection of an appropriate color function within the GSS workflow is an important consideration for effectively communicating semantic similarity on a map. However, a detailed discussion of color function design is outside the scope of this section, as we focus here on the novel methodological challenges of spatial aggregation in GSS.

Use Case: Understanding the Online Perception of Bonn City Districts

This section presents a case study employing GSS to analyze social media data, aiming to understand the character of different city districts. The project sought to identify both specific POIs and broader spatial clusters or AOIs, thereby highlighting the unique attributes of each district for the purpose of informing city marketing and urban planning strategies. This case study demonstrates the potential of GSS for urban analysis, revealing how social media can provide valuable information to both citizens and decision-makers without requiring a-priori knowledge.

Background

Building upon previous work using Instagram data to explore cultural ecosystem services in Bonn’s urban green spaces, the current study addresses existing challenges (Instagrams-Bonn 2020). That prior study found that individuals primarily use these spaces for experiencing nature, relaxation, and emotional expression, with variations observed across different parks and time periods.

While this demonstrated the utility of social media data for urban planning and management, it also highlighted the need for complementary methodologies. A key challenge was the automatic discovery of related posts from a set of keywords. For example, identifying diverse sports activities (e.g., football, frisbee, volleyball) associated with a park without predefined keywords was difficult, further complicated by multilingualism in social media posts.

Although a chi-based spatial binning metric using keyword search was developed (Weckmüller and Dunkel 2023), privacy concerns remained. Subsequently, HyperLogLog was implemented for privacy-aware geospatial search (Weckmüller and Dunkel 2022), providing enhanced privacy but requiring a more complex workflow. These insights and remaining challenges motivated the present case study to design a more efficient and versatile workflow. The aim was to eliminate the need for a-priori knowledge of keywords or the area, address multilingualism, and resolve privacy concerns, ultimately providing an operable tool for the city of Bonn.

Data

This study utilized a dataset of 693,959 public, geotagged Instagram posts from within the defined study area. Data collection and preprocessing involved parsing relevant text captions and Instagram’s precise location coordinates (not requiring geocoding) of each post to create a geo-textual database (Fig. 1). A custom web application was developed to enable interactive GSS of this database. Users could input

keywords or complete queries (e.g., “enjoying a book in a park,” “thriving nightlife”) and visualize the spatial distribution of relevant posts in our online demo on GitHub (<https://do-me.github.io/semantic-hexbins>). To ensure a privacy-sensitive implementation, we made a deliberate choice to omit the original post captions.

Performance and Scalability

Model Choice

Performance and scalability are primarily affected by model selection. In addition, we selected our embedding model based on browser compatibility and technology constraints. Given the rapid pace of development, we used an openly available, pre-trained model rather than training from scratch. To ensure private, offline operation on consumer hardware, we excluded server-client setups and prioritized models compatible with in-browser execution via `transformers.js`.

Microsoft’s “Multilingual E5” family was the top `transformers.js`-compatible multilingual option at the time, with state-of-the-art CLIR, which quantifies the effectiveness of retrieving documents in a target language using queries in a source language, and bitext mining, which assesses the model’s ability to identify corresponding translations (Wang et al. 2024b). For optimal speed and minimal footprint, we chose `multilingual-e5-small` as the smallest variant. Despite its size, it showed competitive CLIR performance: an average normalized discounted cumulative gain at 10 of 60.8 and an average recall at 100 of 92.4 on MIRACL (Zhang et al. 2023), and 93.2% on BUCC 2018, and 64.2% on Tatoeba (Wang et al. 2024b), indicating performance comparable to larger versions.

It is important to note that a new systematic per-language re-evaluation of the chosen model within our specific case study was beyond the scope of demonstrating the GSS workflow itself. Instead, we leverage these established benchmarks and ongoing research in model evaluation, particularly the efforts around the massive multilingual text embedding benchmark (MMTEB by Enevoldsen et al. 2025).

Index Efficiency

Inferencing 693,959 posts completed in under 5 min on an M2 Max chip (fp32 precision). `Multilingual-e5-small`’s 384-dimension vectors are ideal for minimal index size, suitable for static web applications. Storing individual embeddings (~ 550 Mb index) may challenge the RAM of some browsers. Initial experiments with sampling a reduced number of posts showed a linear reduction in index size (e.g., 1000 posts yielded ~0.8 Mb). However, we observed a significant loss of information regarding localized phenomena when using lower post counts (10,000 and 20,000), compared

to samples where all posts were aggregated (visual examples available at <https://github.com/do-me/semantic-hexbins>).

Consequently, we adopted a more effective strategy to balance index size reduction and information preservation: aggregating all embeddings associated with the same location by averaging. This approach resulted in a significant reduction in index size for the entire dataset, which includes 6630 unique Instagram locations and, thus, only 6630 embeddings, compressed to only 5.1 Mb (Table 3).

Using this location-averaged index, we conducted performance evaluations on common consumer-grade devices for single queries. We found that CPU-based inferencing (Intel i7-8550U, AMD Ryzen 7 PRO 6850U, Exynos 9810, A17 Pro) consistently took less than 60 ms. Including subsequent processing steps for a full layer update, such as cosine distance calculations, aggregation using the maximum SAF, sorting, and layer rendering, the total processing time never exceeded 200 ms. Table 3 summarizes the performance characteristics of the index variants tested within our browser-based application. As shown, indices using individual post-level embeddings exhibited inferior visual performance, larger file sizes, and longer processing times. In contrast, averaging embeddings effectively preserves sufficient information for GSS while significantly reducing index size. Remarkably, by averaging embeddings only for locations with more than 99 posts, thereby removing 84.5% of unique locations, we achieved a data file size of only 800 Kb while maintaining comparable visual query results.

Comparing the efficiency of our GSS workflow with traditional GFTS highlights fundamental trade-offs relevant to the findings presented next. While GFTS methods, often leveraging optimizations like specialized indexing structures, excel at rapid lexical retrieval, semantic search using embeddings usually necessitates storing dense numerical vectors. Without optimization, this can lead to substantially larger index sizes (33 Mb vs. 555 Mb, Table 3). However, as demonstrated in our performance analysis, the location averaging strategy employed within our GSS workflow drastically reduces the number of stored vectors, making the resulting file size highly competitive, potentially even smaller than large keyword indices (5.1 Mb vs. 15 Mb).

Regarding query time, due to the additional inferencing step that takes ~33 ms alone using a quantized `multilingual-e5-small` model on an iPhone 15 Pro, unoptimized GSS is usually marginally slower than GFTS. Even though we did not use any search optimization like approximate nearest neighbor (ANN) search or hierarchical navigable small worlds (HNSW), GSS in its version with location-averaged embeddings is already highly compatible to GFTS and within a sub 100 ms range. The key trade-off remains: keyword systems are efficient for lexical retrieval, while GSS provides efficiency for semantic retrieval, with aggregation making its storage requirements practical.

Table 3 Data file size (gzipped JSON) and end-to-end query time for individual post and aggregated location embeddings using GSS including ~33 ms inferencing time with and query time for individual

texts using GFTS on an iPhone 15 Pro. Number of unique embeddings in bold. Code on <https://github.com/do-me/semantic-hexbins>

Search variant	Posts	Locs	Constraint	Size [Mb]	Time [ms]
GSS indiv. embs	693,959	6630	N/A	555	N/A
GSS indiv. embs	40,000	4045	N/A	32.1	131
GSS indiv. embs	20,000	3157	N/A	16	92
GSS indiv. embs	10,000	2359	N/A	8	72
GSS loc.-av. embs	693,959	6630	N/A	5.1	68
GSS loc.-av. embs	693,293	5964	> 1 post/location	4.6	65
GSS loc.-av. embs	686,558	4487	> 9 posts/location	3.5	58
GSS loc.-av. embs	542,722	1028	> 99 posts/location	0.8	42
GFTS indiv. texts	693,959	6630	N/A	33	91
GFTS loc.-agg. texts	693,959	6630	N/A	15	33
GFTS topic subset	39,247	2643	Subset for “nature”	0.048	N/A

Findings

Applying the GSS workflow to the Bonn Instagram dataset revealed distinct semantic profiles for each city district, effectively capturing dominant themes and activities. The analysis successfully differentiated districts based on their online perception. For example, the city center was characterized by posts related to historical landmarks and cultural events, while the Altstadt district showed a strong association with vibrant nightlife, and the Rheinaue area reflected a focus on nature and recreation. These findings aligned with general knowledge of Bonn’s districts, demonstrating the GSS workflow’s ability to capture meaningful semantic distinctions from social media data. Furthermore, the analysis captured temporal dynamics within district character. For instance, fitness-related posts were prevalent throughout the day across various districts, while shisha bar mentions were concentrated in the evenings, particularly in the Altstadt. This temporal sensitivity highlights the potential of GSS to monitor time-dependent urban activity patterns. Localized phenomena were also readily identifiable. A prominent springtime floral display, for example, became visually apparent in the Altstadt district during daytime searches (Fig. 4), reflecting real-world events captured in social media.

To validate these findings, we employed several approaches. Firstly, we performed a comparative assessment of GSS outputs against results from a conventional keyword-based search application for Bonn (<https://github.com/do-me/hexbins>). This assessment was conducted in consultation with personnel from the City of Bonn involved in urban planning and marketing, serving as domain experts and representatives of the intended user group. The process involved presenting GSS-generated thematic hotspots for specific queries (e.g., “Cherry blossom”) alongside outputs from keyword searches for equivalent terms. Key thematic

hotspots identified by GSS showed strong spatial correlation with the primary areas highlighted by keyword searches. This expert feedback indicated consistency in identifying relevant geographic areas while also highlighting enhanced flexibility through semantic understanding, which was perceived as beneficial by the city personnel.

Secondly, we systematically inspected public Instagram posts from top-ranking GSS-identified locations. The GSS prototype facilitated this by enabling browser navigation from high-scoring hexagonal bins to the public Instagram pages of associated locations. Manual review of the publicly displayed top posts at these locations confirmed their semantic relevance to the queries (e.g., “Cherry blossom” in Altstadt, Fig. 4), a finding corroborated by City of Bonn personnel based on their domain expertise, thus validating the GSS workflow’s accuracy in linking semantic queries to appropriate geographic areas.

From a usability perspective, evaluation of the prototype application yielded feedback regarding visualization and SAFs. User testing within the Bonn case study context indicated that hexagonal binning served as an effective visualization method for interactively exploring data at various scales. Experimentation confirmed the significant impact of SAF choice (Table 2) on insight generation. While functions like minimum or mean proved useful for specific analytical tasks, participant feedback and observation suggested the maximum SAF provided the most stable and intuitively interpretable results for general exploration tasks in this study across different zoom levels (i.e., different effective spatial resolutions of the hexagonal grid, as illustrated in Fig. 4). Users noted that the maximum function consistently highlighted areas containing highly relevant documents across various zoom levels, effectively mitigating the MAUP-driven “value dilution” issue considered during design (“Usability Considerations: from User Query to Spatial Visualization” section).

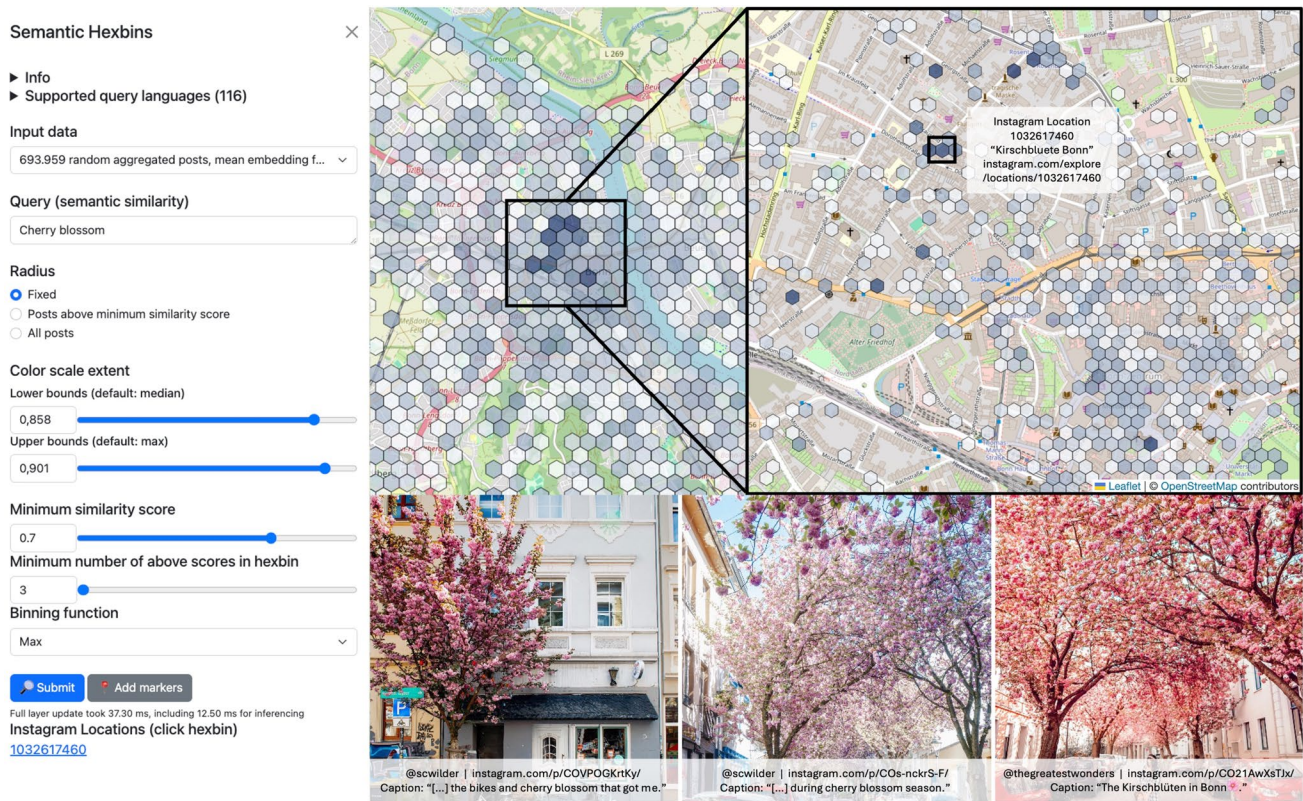


Fig. 4 Web interface with query results for “Cherry blossom” and three top-ranking public posts for the Instagram location displayed in the interactive web app for Bonn with default settings like maximum as SAF. Dark blues indicating the hotspot in the Altstadt district

The robustness of the maximum SAF to changes in scale is crucial for interactive hotspot discovery. It ensures that if a bin, regardless of its current size due to zoom level, contains even one highly relevant document, its presence is signaled. This prevents strong local semantic signals from being averaged out or obscured as the aggregation scale changes—a common concern with MAUP (Openshaw 1984; Andrade et al. 2020). Thus, users can confidently zoom from a general overview to finer details, knowing that significant semantic concentrations are less likely to be missed due to aggregation artifacts inherent in other SAFs. This SAF facilitated clearer exploration of both specific POIs and broader AOIs. The interactive visualization, allowing adjustment of SAFs and color scales (Fig. 4), proved integral to the iterative search process. Adjusting color scale bounds based on similarity value distribution (e.g., median and maximum) provided a useful starting point for refinement. While these qualitative validation steps and user feedback support the GSS workflow’s effectiveness, the lack of standardized quantitative benchmarks remains a limitation. Future work should focus on developing such benchmarks for more rigorous evaluation.

remain consistent across zoom level 13 (main window) and 16 (bottom right) instead of showing a “value dilution” effect. Designed for expert usage to explore different SAF and visualization settings

Discussion

Strengths and Versatility

The GSS methodology presented in this study offers several advantages, positioning it as a robust tool for analyzing geo-textual data. Its key strength lies in its use of semantic search rather than traditional keyword-based approaches, enabling users to retrieve and link information based on meaning rather than exact word matches. As previously noted, researchers and analysts frequently lack precise a-priori keyword knowledge and thus need a workflow facilitating easier data exploration. The findings from the Bonn case study demonstrate the practical utility of the GSS workflow for urban analysis. It provided actionable insights into Bonn’s online image and district characteristics, which were disseminated via internal reports to inform city marketing and urban planning strategies.

A fundamental benefit of our GSS workflow is its adaptability, achieved using current multilingual embedding models. This capability is critical when working with large, multilingual geo-textual datasets, such as those derived from social media, where users often employ multiple languages concurrently

and even mix them with slang. Embedding models effectively address this challenge, in contrast to keyword-based searches, which tend to underperform in such contexts. The modular nature of our workflow enables the easy exchange and adaptation of different components for various use cases. For instance, instead of geo-textual data, geo-visual data could be used with the same embedding model (e.g., bge-M3, Luo et al. 2024), aligning both text and images within the same vector space for improved CLIR quality and a broader application spectrum.

Furthermore, future models, such as specialized models for social media data, underrepresented languages, or similar can be easily integrated in a plug-and-play manner. Beyond embeddings, the GSS workflow can be adapted for other tasks, such as sentiment analysis, without the embedding generation or distance calculation steps. Like HyperLogLog-based workflows (Weckmüller and Dunkel 2022), embeddings are suitable for geo-textual data workflows requiring high compression rates through aggregation, for instance in lean frontend applications. Further compression could be achieved via binary and scalar quantization (Shakir et al. 2024) and Matryoshka embeddings (Kusupati et al. 2022). Adding to these strengths, our case study demonstrates that the GSS workflow can fully omit the original document texts following inferencing, thereby providing a notable privacy advantage compared to keyword-based methods that necessitate storing the original textual data.

Challenges and Limitations

Despite the demonstrated strengths and versatility of the GSS methodology, it is important to recognize several inherent challenges and limitations. Many of these stem from the underlying nature of text embeddings and the complexity of semantic search itself.

A primary challenge is to determine appropriate similarity score thresholds for relevance. Semantic similarity scores derived from cosine distance or similar metrics provide a continuous spectrum of relatedness rather than a binary classification of relevant or irrelevant. As a result, establishing a definitive cutoff point to effectively filter out unrelated documents becomes a non-trivial challenge. Every query, even for semantically distant concepts, will yield a similarity score, making it difficult to objectively distinguish truly relevant results from those that are only marginally related or completely spurious. This lack of clear semantic boundaries in the embedding space requires careful consideration and potentially query-dependent adjustment of similarity thresholds, adding complexity to the practical application of GSS.

Furthermore, while the flexibility offered by SAFs is a methodological strength, it also introduces usability complexities. The wide variety of SAFs, each with its own advantages and disadvantages (as detailed in Table 2), can be overwhelming to users, especially those without expertise in spatial analysis or semantic search. Choosing the most

appropriate SAF for a given query and analytical objective requires a nuanced understanding of their respective behaviors and potential impact on visualization and interpretation. This inherent flexibility therefore presents a usability challenge, potentially hindering accessibility for non-expert users and requiring careful guidance or automated support mechanisms.

A major conceptual limitation is the “black box” nature of current embedding models, a challenge central to the discourse on responsible GeoAI (Marasinghe et al. 2024). These models, often based on deep neural networks, operate on complex, high-dimensional representations of meaning that are not easily interpreted by humans. This lack of transparency can make it difficult to understand why certain documents are considered semantically similar or dissimilar, hindering the debugging process and potentially undermining user confidence in the system’s output. The inherent opacity of embedding models makes it difficult to fully validate and explain the semantic relationships captured and visualized by GSS, limiting the interpretability of results and requiring cautious interpretation of findings. As research on the interpretability of embedding models is still in its infancy (Templeton et al. 2024), this “black box” characteristic remains a significant limitation for practical applications that require transparency and explainability.

Ethical considerations regarding the biases embedded in pre-trained language models are also paramount. Extensive research has documented the presence of societal biases, including ethnic stereotypes, racism, sexism, and Eurocentrism, in widely used embedding models (Petreski and Hashim 2023). These biases can be inadvertently propagated and even amplified when such models are used to analyze geo-textual data, potentially leading to distorted representations of urban perceptions and reinforcing existing societal inequalities. In the context of urban studies, where data-driven insights can inform policy decisions and resource allocation, the potential for biased GSS results to perpetuate inequitable or discriminatory outcomes is a serious ethical concern that requires careful mitigation strategies and critical awareness.

It is crucial to recognize the inherent privacy limitations of text embeddings. While initially hypothesized to provide privacy-preserving properties (Abdalla et al. 2020) by reducing text to an abstract N-dimensional latent space, studies have shown that embeddings are not completely anonymizing. Research has demonstrated the feasibility of reconstructing substantial portions of the original text from embeddings (Morris et al. 2023; Zhu et al. 2024), raising concerns about the privacy implications of using and storing embedding representations of user-generated content, especially in the context of sensitive geo-textual data. Given the increased vulnerability of geoprivacy (Kebler and McKenzie 2018), especially when location information is combined with potentially reconstructable textual content,

robust privacy-preserving techniques are essential to mitigate these risks in GSS applications.

Finally, rigorously evaluating semantic search against traditional keyword-based approaches, particularly within a geospatial context, remains challenging due to the limitations of existing benchmarks for geo-textual search (Chen et al. 2021). While semantic similarity evaluation has advanced with widely adopted public benchmarks such as BEIR (Thakur et al. 2021) and MTEB (Muennighoff et al. 2022), along with its extensive multilingual counterpart MMTEB (Enevoldsen et al. 2025) which itself underscores performance variations and a general skew in model attention towards higher-resource languages, these have emerged only recently and are already the subject of ongoing debate regarding their scope and effectiveness, reflecting the difficulties in standardizing semantic evaluation.

Critically, these benchmarks primarily focus on textual retrieval tasks and do not adequately address the unique aspects of geospatial relevance and map-based visualization inherent to GSS. Furthermore, establishing purely objective indicators to validate GSS findings faces significant challenges due to the inherent subjectivity in assessing semantic relevance and perceived quality within complex domains like urban studies.

Objectively quantifying the expressive and intuitive aspects of human perception and language use is fundamentally problematic (Dakin 2003). As Wherrett (1998) argues, even seemingly objective quantitative metrics or algorithmic assessments can disguise underlying subjectivity, given that criteria selection and weighting often stem from subjective decisions. Consequently, defining and universally agreeing upon truly objective metrics for the “correctness” or “quality” of GSS visualizations remains a significant hurdle. This necessitates the continued use of qualitative validation and user feedback as crucial, albeit inherently subjective, components of evaluation in the current state of the field.

Future Research Directions

Building on the identified challenges and limitations, several directions for future research emerge. First, given the fundamental role of model embedding in GSS, any advances in the quality and capabilities of these models will directly translate into improvements in GSS performance and accuracy, including multimodal capabilities. Exploration of fine-tuning strategies for embedding models, specifically tailored to the CLIR and urban data domains, could provide significant performance gains. Further research is needed to improve the interpretability and explainability of embedding models, allowing a deeper understanding of their learned features and facilitating the interpretation of results.

Similarly, addressing the ethical considerations related to biases in embedding models is paramount and requires

the development and implementation of effective mitigation strategies. Given the geoprivacy vulnerabilities, future work should prioritize the exploration of robust privacy-preserving techniques within GSS workflows, possibly incorporating privacy budgets (Ma et al. 2024) or differential privacy methods. Developing methods to automatically recommend appropriate SAFs based on user queries is critical to improving usability and optimizing search results. To further enhance data exploration capabilities, future work could explore the integration of embedding-based topic clustering techniques. Methods such as UMAP, PCA, t-SNE, or HUMAP (Marcílio-Jr et al. 2024) combined with clustering algorithms such as k-means could be explored to identify spatial thematic clusters within geo-textual datasets using the embeddings generated by the GSS workflow.

Hybrid search strategies that combine semantic search with traditional full-text search warrant further investigation as a potential means of balancing precision and recall and improving the overall search experience, possibly using reranking methods such as BM25 or specialized reranking models. Finally, the establishment of standardized evaluation frameworks designed specifically for GSS is essential to objectively compare different GSS approaches and drive progress in the field, going beyond general semantic benchmarks such as BEIR and MTEB to include geospatial relevance and usability metrics.

Conclusion

We present a novel geospatial semantic search (GSS) workflow that leverages multilingual text embeddings for improved search and exploration of geo-textual data. Our workflow specifically addresses the challenges within urban studies applications and systematizes the process into three distinct steps. Using GSS, we directly address the limitations of traditional keyword-based search by focusing on semantic similarity. This paradigm shift enables information retrieval based on meaning and conceptual relatedness. It is critical to effectively handle the ambiguity, multilingualism, and dynamic nature of user-generated content that typically hampers keyword-based methods.

The core innovation to improve geo-textual search is the aggregation of text embeddings for geographic entities and the comparison of these aggregated representations with user queries. This approach facilitates efficient semantic search and exploration of large datasets. It enables the identification of spatial patterns and thematic trends relevant to urban phenomena that are often missed by keyword methods. In addition, the aggregation process significantly reduces index size, improving scalability and performance, especially for client-side applications, while offering potential privacy benefits by omitting original text.

Our case study in Bonn, Germany, demonstrated the practical utility of GSS for urban analysis. The workflow successfully revealed distinct semantic profiles for neighborhoods, providing actionable insights for urban planning and marketing. It demonstrated the power of semantic search combined with interactive map visualization in a real-world geospatial context. The study highlighted the effectiveness of hexagonal binning and confirmed maximum SAF as an intuitive choice for general exploration, demonstrating how visualization choices and analysis context can be used to enhance usability.

The modular design of the GSS workflow offers significant advantages in adaptability and extensibility. It allows for the integration of diverse data sources, evolving embedding models, alternative aggregation methods, and different visualization techniques. While acknowledging challenges such as similarity threshold determination, the interpretability limitations of embedding models, embedded biases, and geoprivacy concerns, GSS provides a robust and versatile tool for urban analysis. It unlocks the semantic richness of geo-textual data, providing enhanced search accessibility, improved search quality, and scalable performance in geospatial contexts. This makes it particularly useful for urban studies and related planning fields that require effective exploration of complex geo-textual data with sensitivity to subtle differences in meaning.

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Data Availability The source code, demonstration application, and data supporting the findings of this study are openly available on GitHub at <https://github.com/do-me/semantic-hexbins>. The specific version associated with this publication is archived on Zenodo and can be cited as: Weckmüller, D. (2025). Semantic Hexbins: Software and Data for Geospatial Semantic Search. Zenodo. <https://doi.org/10.5281/zenodo.15213530>.

Declarations

Ethical Approval This work complies with all applicable ethical standards and does not contain any studies with human participants or animals performed by any of the authors. The data used for the demo of this study was collected from publicly available Instagram posts.

Consent to Participate The manuscript does not report a study involving human participants.

Conflict of Interest The authors declare no competing interests.

Data Protection The authors have taken necessary steps to protect the privacy of individuals whose data was used in this study. The data was anonymized and aggregated to prevent identification of individual users and respect geoprivacy.

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